

lab_tour

DAY 2 · one continuous walking shot · quiet room

- **Every rule gets made at the object it belongs to** — not batched at a bench
- **Name every station out loud as you reach it** — nothing else reaches a caption
- Route and the lines to say are on the **Safety & Lab Tour labsheet**
- Door → three bays → safety corner → media/microwave → gels → PCR
- → miniprep → flex → open bench → incubator → -80 → transformation → hood → sterile bench
- **Do not pull the safety shower.** Point, say what it is for, move on
- **EtBr station needs your wording** — the labsheet leaves it blank on purpose

What control strains are and why we keep them

ctrl_intro

DAY 2 • no bench work

- A **control strain** is a strain of known resistance, kept so a new batch of plates can be trusted
- Four of them: **carb • kan • spec • erm**
- They live in the **–80 box — supervisors only**. They are the irreplaceable material in the room
- Pour a batch, streak the controls, and the plate either selects or it does not
- Sets up the streak and transformation clips already shot

Examine and photograph plates

ctrl_read

DAY 2 · the overnight plates

- **Say every judgement out loud** — all of this is visual
- Name the sector that grew, and the ones that did not
- Sterility plates: contamination in the agar, bottle or hood
- Kanamycin — **rescued vs the no-rescue half**
- Cells-only plates should be empty
- Photograph · record batch date, who poured, result

parafilm

DAY 2 · the plates you are keeping

- **These plates are being held for picking later** — the seal is what makes that possible
- Cut a **1-inch strip along the long edge** of the roll
- Hold against the lid rim · stretch until it grips · **spin the plate**, keeping tension
- Press the end down onto itself to seal
- **Culture Fridge (4 °C)** — the one for cultures and plates that have cells on them
- **Not the Main / Cell-Free fridge**, which is sterile dishes, DMEM and gels
- Store **inverted** · stable **1–2 weeks**; get back to them within the week
- **Never parafilm anything going into the incubator** — cells need air

Pick BestP colonies into a block

bestp_pick

DAY 3 · full 24-well block

- **Photograph the plates under blue light first** — the only record of what you picked from
- **4 mL 2YT + carb** per well
- **4 colonies per plate** · block is **4 × 6**
- **Left to right, in the sample order** — otherwise the data entry scrambles silently
- Airpore sheet on, then cover
- Multitron overnight
- Parafilm the plates · upside-down in the fridge

Pick pP6 clones into a 15 mL conical

pp6_pick

DAY 3 · do 2

- Brightest green under blue light · **mark them on the plate**
- **Small and slow-growing can be the strongest** — do not just take the biggest
- Well-isolated only · watch for **satellites**, the white specks around a green colony
- **4 mL 2YT + carb**
- Sterile toothpick · **drop it in and leave it there**
- Shaking incubator overnight
- Parafilm the plate · upside-down in the fridge

Tecan the BestP block

bestp_tecan

DAY 4 • Tecan M Nano • 48 read wells

- Check the cultures grew and are fluorescent **before anything else**
- **100 µL** per culture into a **black-walled** 96-well plate
- P200 multichannel • **columns of 8** • 2 draws per sample
- Fluorescence: **Ex 483 / Em 525**, bandwidths 20, **gain 40**, 20 reads
- high sensitivity • 40 µs • Z 11019 µm • 30 °C • COS96ft
- **Also read OD600** — fluorescence alone is uninterpretable
- Save to the **USB stick**; that computer has no internet
- **Remove your plate before shutdown** • bleach and rinse it

What BestP is measuring

bestp_intro

WHENEVER · no bench work

- **RPU** = fluorescence per OD unit, normalised to **pJ01 = 1**
- **pJ12** very weak · **pJ01** the reference · **pJ19** very strong
- Same backbone, same reporter — so **only the promoter differs**
- Raw fluorescence is not comparable across runs; RPU is
- The goal: pP6 clones that **match or beat pJ19**